what light means

digitalisierung und programmierung · prof. dr. nicolas meseth

1. agreement
2. counting information
3. your alphabet

part 1

agreement

what makes light mean something?

image: ai-generated (gpt-image-2)

a symbol is an agreed, distinguishable state.

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distinguishable: the receiver can tell it apart from the others.

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agreed: both sides gave it the same meaning, beforehand.

the sender's table

red	→	a
green	→	b
blue	→	c
yellow	→	d

the receiver's table

sent: → abca

the sender's table

red	→	a
green	→	b
blue	→	c
yellow	→	d

sent: → abca

the receiver's table

red	→	b
green	→	c
blue	→	d
yellow	→	a

read as: bcdb

every colour arrived correctly. the message did not.

the agreement, written down

```
ALPHABET = {
    "red":    "00",
    "green":  "01",
    "blue":   "10",
    "yellow": "11",
}
```

both sides need this table. it is not data, it is a promise.

part 2

counting information

the guessing game, measured

possibilities left

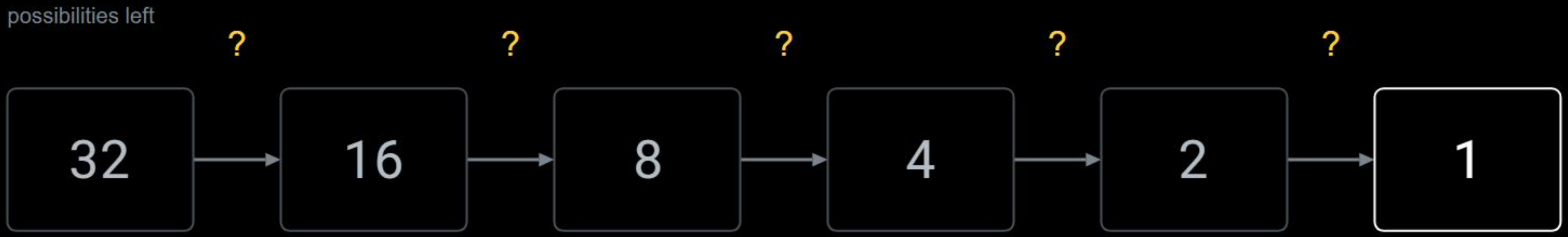
32

the guessing game, measured

possibilities left

$$32 \xrightarrow{?} 16 \xrightarrow{?} 8 \xrightarrow{?} 4 \xrightarrow{?} 2 \xrightarrow{?} 1$$

the guessing game, measured



each question halves what is left

five questions. every time.

not every answer is worth one bit

"is it red?"		
yes	→	16 left
no	→	16 left

always exactly one halving

"is it the ace of spades?"		
yes	→	0 left, solved
no	→	31 left

almost always the second line

you cannot choose which answer you get.

what a question is worth on average

question	answer	chance	you learn	on average
"is it red?"	yes	1/2	1 bit	1 bit
	no	1/2	1 bit	
"is it the ace of spades?"	yes	1/32	5 bit	0.2 bit
	no	31/32	0.05 bit	

a rare answer tells you a lot. an expected answer tells you little.

and in general

in words

how likely an answer is × what that answer is worth, added up

in symbols

$$E = \Sigma\, p_i \cdot \log_2(1/p_i)$$

p_i	how likely that answer is
$\log_2(1/p_i)$	what that answer is worth, in bits

and in general

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the two questions from before

$0.5 \cdot 1 + 0.5 \cdot 1 = 1$ bit

$0.03 \cdot 5 + 0.97 \cdot 0.05 = 0.2$ bit

a fair split is the only way to get a whole bit.

0	1

one bit = one decision between two possibilities

a light switch holds one. no computer required.

0

1

one bit = one decision between two possibilities

$H = \log_2(N)$ uncertainty: how many halvings are still missing

$I = H_1 - H_2$ information: what an answer removed

32 cards are not the uncertainty. the 5 is.

a light switch holds one. no computer required.

try it: which question is worth more?

ask about… — or click a face to guess. count the faces before you ask: which question splits them best?

glasses?	a hat?
a beard?	curly hair?
earrings?	a bow tie?
grey hair?	blonde hair?
dark hair?	female?

information collected: **0.00** of **4.00** bits

faces left	questions asked	bits per question	needed in total
16	0	–	4.00
of 16	a guess counts too	perfect halving: 1.00	$\log_2(16)$ bits

what a bigger alphabet buys

colours	bits per symbol
2	$\log_2(2) = 1$
4	$\log_2(4) = 2$
8	$\log_2(8) = 3$
16	$\log_2(16) = 4$

what a bigger alphabet buys

colours	bits per symbol	time for the same file, at the same symbol rate
2	$\log_2(2) = 1$	100 %
4	$\log_2(4) = 2$	50 %
8	$\log_2(8) = 3$	33 %
16	$\log_2(16) = 4$	25 %

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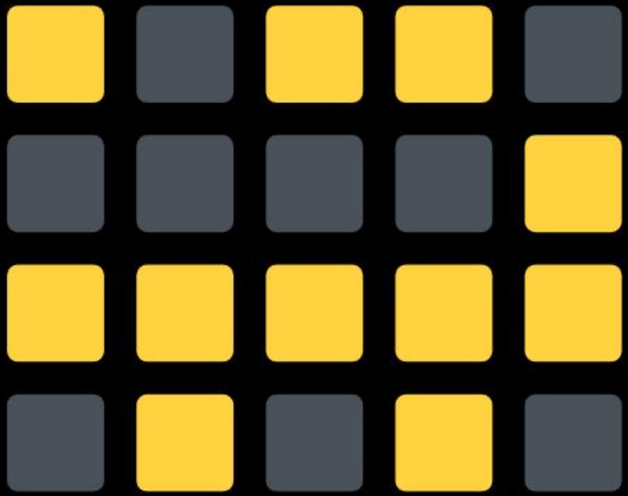
same file, same symbol rate. only the alphabet changed.

two symbols, five flashes per character



10110

00001

11111

01010

$$2^5 = 32$$

32 combinations: room for a whole alphabet

what a big alphabet does per symbol, two symbols do in a group.

when symbols are not equally likely

e • most common letter

t —

q — — • — rare letter

in real text, symbols are not equally likely

rare symbols may cost more. common ones must not.

part 3

your alphabet

the payoff

more bits per symbol

the same file needs fewer symbols

and arrives sooner

the price

the colours move closer together

noise reaches across the gap

and symbols are read wrong

the size of your alphabet is not a matter of taste. it is a measurement.

in the workshop

agree on your alphabet and write it down
measure a profile for every symbol
fill send_symbol() and receive_symbol()
run a whole word and count the errors

light means nothing
until you agree what it means.

light means nothing
until you agree what it means.

and how much it means, you can count.